Hydrogenic Atom

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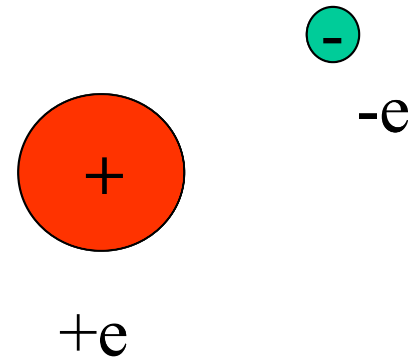
We assume that the potential energy is given by Coulomb's Law



$$V(r) = \int_{\infty}^{r} \frac{e^2}{4\pi\varepsilon_0} \frac{dr}{r^2} = -\frac{e^2}{4\pi\varepsilon_0 r}$$

Let's look at this solution in more detail as it will help illuminate what real atoms behave like.

Wave Equation

- Starting with the radial wave equation

$$\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left\{\frac{2\mu r^2}{\hbar^2}\left[\varepsilon_{\text{int}}-V_{\text{int}}(r)\right]-\alpha\right\}R=0$$

- Inserting V and α (the two particle separation constant)

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)-\left\{\frac{l(l+1)}{r^2}+\frac{2\mu}{\hbar^2}\left[\varepsilon_{\text{int}}+\frac{e^2}{4\pi\varepsilon_0 r}\right]\right\}R=0$$

Solution

- It is usual to transform the equation so that

$$\frac{1}{\rho^2}\frac{d}{d\rho}\left(\rho^2\frac{dR}{d\rho}\right)-\left[\frac{l(l+1)}{\rho^2}-\frac{n}{\rho}+\frac{1}{4}\right]R=0$$

- where

$$\rho=\frac{2r}{na_0}\qquad n^2=-\frac{\hbar^2}{2\mu a_0^2\varepsilon_{el}}\qquad a_0=-\frac{\hbar^2}{\pi\mu e^2}$$

This is a Laguerre equation

Solution

- The power series solution is

$$R_{nl}(\rho) = -\left\{\frac{(n-l-1)!}{2n[(n+l)!]^3}\right\}^{1/2}\left(\frac{2}{na_0}\right)^{3/2}\rho^l \exp(-\rho/2)L_{n+1}^{2l+1}(\rho)$$

- Where

$$L_{n+1}^{2l+1}(\rho) = \sum_{k=0}^{n-l-1}(-1)^{k+1}\frac{[(n+l)!]^2}{(n-l-1-k)!(2l+1+k)!k!}\frac{r^k}{}$$

This is the associated Laguerre polynomial

Principal Quantum Number

- The principal quantum number n can take on the values

$$n = 1, 2, 3, \ldots, l < n$$

- thus

$$l = 0, 1, 2, \ldots, n - 1$$

Radial Wave Functions

Table 6.3 *The first few associated Laguerre polynomials and radial functions for the hydrogen atom*

$n = 1, l = 0$	$L_1^1(\rho) = -1$	$R_{10} = \left(\frac{1}{a_\circ}\right)^{3/2} 2\exp\left(-\frac{r}{a_\circ}\right)$
$n = 2, l = 0$	$L_2^1(\rho) = -2!\,(2-\rho)$	$R_{20} = \left(\frac{1}{2a_\circ}\right)^{3/2} 2\left(1-\frac{r}{2a_\circ}\right)\exp\left(-\frac{r}{2a_\circ}\right)$
$n = 2, l = 1$	$L_3^3(\rho) = -3!$	$R_{21} = \left(\frac{1}{2a_\circ}\right)^{3/2} \frac{2}{\sqrt{3}}\left(\frac{r}{2a_\circ}\right)\exp\left(-\frac{r}{2a_\circ}\right)$
$n = 3, l = 0$	$L_3^1(\rho) = -3!(3-3\rho+\frac{1}{2}\rho^2)$	$R_{30} = \left(\frac{1}{3a_\circ}\right)^{3/2} 2\left[1-\frac{2r}{3a_\circ}+\frac{2}{3}\left(\frac{r}{3a_\circ}\right)^2\right]\exp\left(-\frac{r}{3a_\circ}\right)$
$n = 3, l = 1$	$L_4^3(\rho) = -4!\,(4-\rho)$	$R_{31} = \left(\frac{1}{3a_\circ}\right)^{3/2} \frac{4\sqrt{2}}{3}\left(\frac{r}{3a_\circ}\right)\left(1-\frac{r}{6a_\circ}\right)\exp\left(-\frac{r}{3a_\circ}\right)$
$n = 3, l = 2$	$L_5^5(\rho) = -5!$	$R_{32} = \left(\frac{1}{3a_\circ}\right)^{3/2} \frac{2\sqrt{2}}{3\sqrt{5}}\left(\frac{r}{3a_\circ}\right)^2\exp\left(-\frac{r}{3a_\circ}\right)$

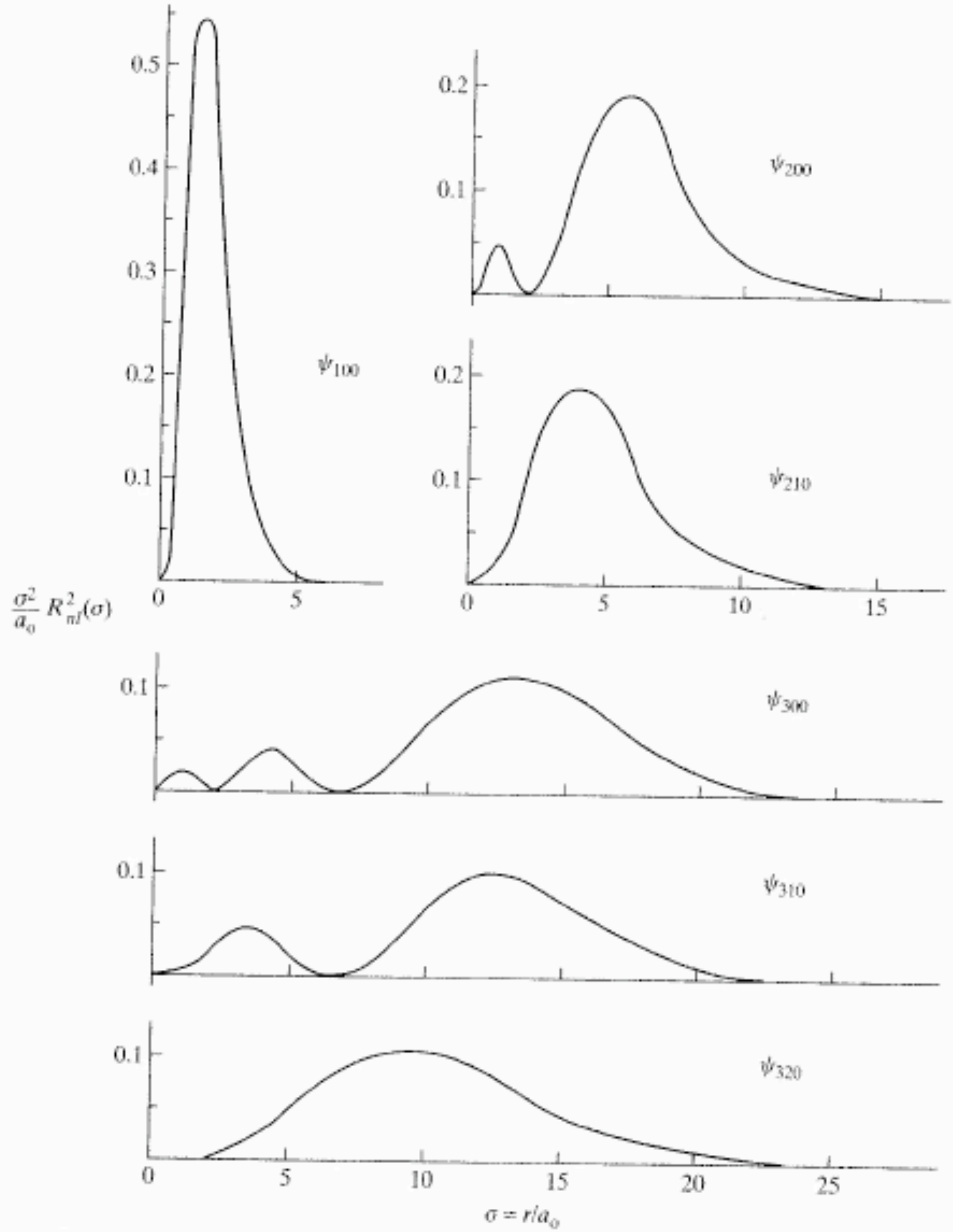


Figure 6.4 The dimensionless probability density function, $\sigma^2 R^2_{nl}(\sigma)/a_\circ$, associated with the radial portion of various wave functions for atomic hydrogen, where $\sigma = r/a_\circ$.

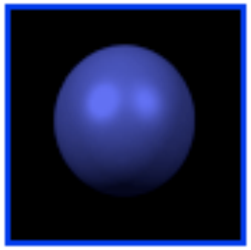
Total Wave Function

$$\psi_{nlm}(r,\theta,\phi) = R_{nl}(\rho)Y_l^m(\theta,\phi)$$

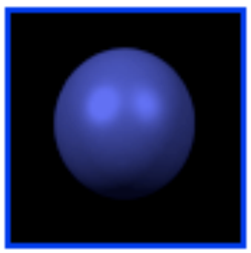
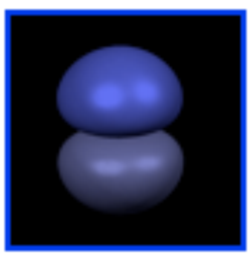
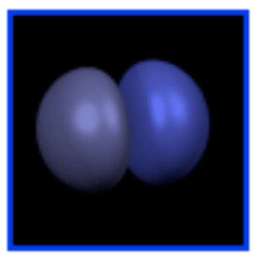
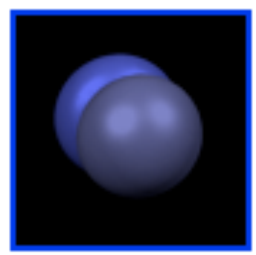
Table 6.4 *The wave functions for atomic hydrogen, including hydrogen-like species with Z protons* ($n = 1, n = 2, n = 3$)

$n = 1, l = 0, m = 0$	$\psi_{100} = \frac{1}{\sqrt{\pi}}(\frac{Z}{a_\circ})^{3/2}\exp(-Zr/a_\circ)$
$n = 2, l = 0, m = 0$	$\psi_{200} = \frac{1}{4\sqrt{2\pi}}(\frac{Z}{a_\circ})^{3/2}[2 - (Zr/a_\circ)]\ \exp(-Zr/2a_\circ)$
$n = 2, l = 1, m = 0$	$\psi_{210} = \frac{1}{4\sqrt{2\pi}}(\frac{Z}{a_\circ})^{3/2}(Zr/a_\circ)\exp(-Zr/2a_\circ)\cos\theta$
$n = 2, l = 1, m = \pm 1$	$\psi_{211} = \frac{1}{8\sqrt{\pi}}(\frac{Z}{a_\circ})^{3/2}(Zr/a_\circ)\exp(-Zr/2a_\circ)\sin\theta\, e^{\pm i\phi}$
$n = 3, l = 0, m = 0$	$\psi_{300} = \frac{1}{81\sqrt{3\pi}}(\frac{Z}{a_\circ})^{3/2}[27 - 18(Zr/a_\circ) + 2(Zr/a_\circ)^2]\exp(-Zr/3a_\circ)$
$n = 3, l = 1, m = 0$	$\psi_{310} = \frac{\sqrt{2}}{81\sqrt{\pi}}(\frac{Z}{a_\circ})^{3/2}[6(Zr/a_\circ) - (Zr/a_\circ)^2]\exp(-Zr/3a_\circ)\cos\theta$
$n = 3, l = 1, m = \pm 1$	$\psi_{311} = \frac{1}{81\sqrt{\pi}}(\frac{Z}{a_\circ})^{3/2}[6(Zr/a_\circ) - (Zr/a_\circ)^2]\exp(-Zr/3a_\circ)\sin\theta\, e^{\pm i\phi}$
$n = 3, l = 2, m = 0$	$\psi_{320} = \frac{1}{81\sqrt{6\pi}}(\frac{Z}{a_\circ})^{3/2}(Zr/a_\circ)^2\exp(-Zr/3a_\circ)(3\cos^2\theta - 1)$
$n = 3, l = 2, m = \pm 1$	$\psi_{321} = \frac{1}{81\sqrt{\pi}}(\frac{Z}{a_\circ})^{3/2}(Zr/a_\circ)^2\exp(-Zr/3a_\circ)\sin\theta\cos\theta\, e^{\pm i\phi}$
$n = 3, l = 2, m = \pm 2$	$\psi_{322} = \frac{1}{162\sqrt{\pi}}(\frac{Z}{a_\circ})^{3/2}(Zr/a_\circ)^2\exp(-Zr/3a_\circ)\sin^2\theta\, e^{\pm 2i\phi}$

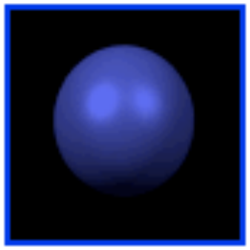
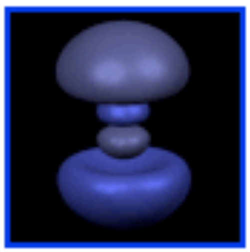
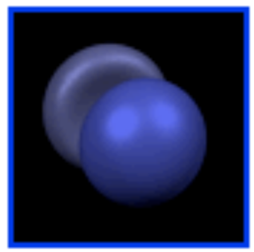
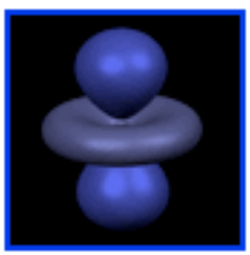
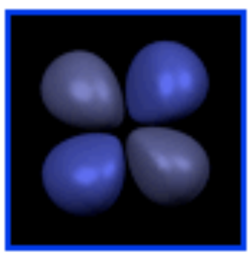
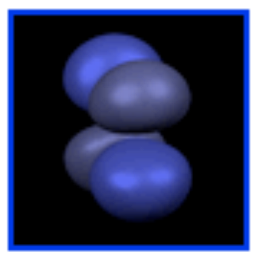
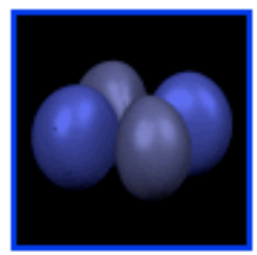
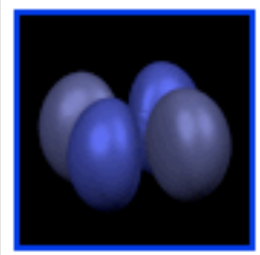
- **n=1:**

	m=0
l=0	

- **n=2:**

	m=0	m=1 Re	m=1 Im
l=0			
l=1			

- **n=3:**

	m=0	m=1 Re	m=1 Im	m=2 Re	m=2 Im
l=0					
l=1					
l=2					

Energy Levels

- The energy becomes

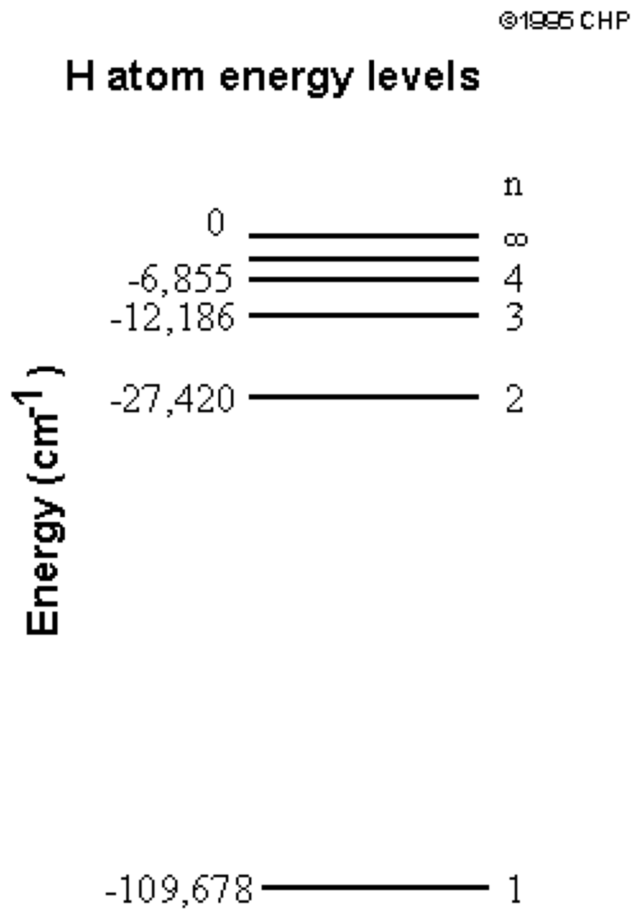
$$\varepsilon_n = -\frac{Z^2 e^4 \mu}{32\pi^2 \varepsilon_0^2 \hbar^2} \frac{1}{n^2} \quad , \qquad n = 1,2,3,...$$

- Note that for this simple model the energy is only a function of n.

Hydrogen Atom

This is illustrated by the energy levels of the hydrogen atom.

When *n* becomes large, the electron separates from the nucleus and we say the atom has ionized.



Degeneracy

For this ideal "hydrogenic" atom, the other quantum numbers, l and m_l, do not affect the energy. However, their values are limited by the value of n. One can show that allowed values of l and m_l are limited as

$$l = 0,1,2,3,\ldots,n-1 \qquad l = i + |m_l| \;, \quad i = 0,1,2,\ldots$$

So that

$$g_n = n^2$$

Degeneracy

One further complication involves the electrons. Quantum mechanics tells that electrons, in addition to orbiting the nuclei, have spin. The spin angular momentum is given by

$$S^2 = s(s+1)\hbar^2$$

Where s is the spin angular momentum quantum number. The value of s is limited to 1/2. As with the general case,

$$S_z = m_s\hbar$$

where

$$m_s = \pm\frac{1}{2}$$

Degeneracy

Therefore, for our simple hydrogenic atom the total degeneracy is

$$g_n = 2n^2$$